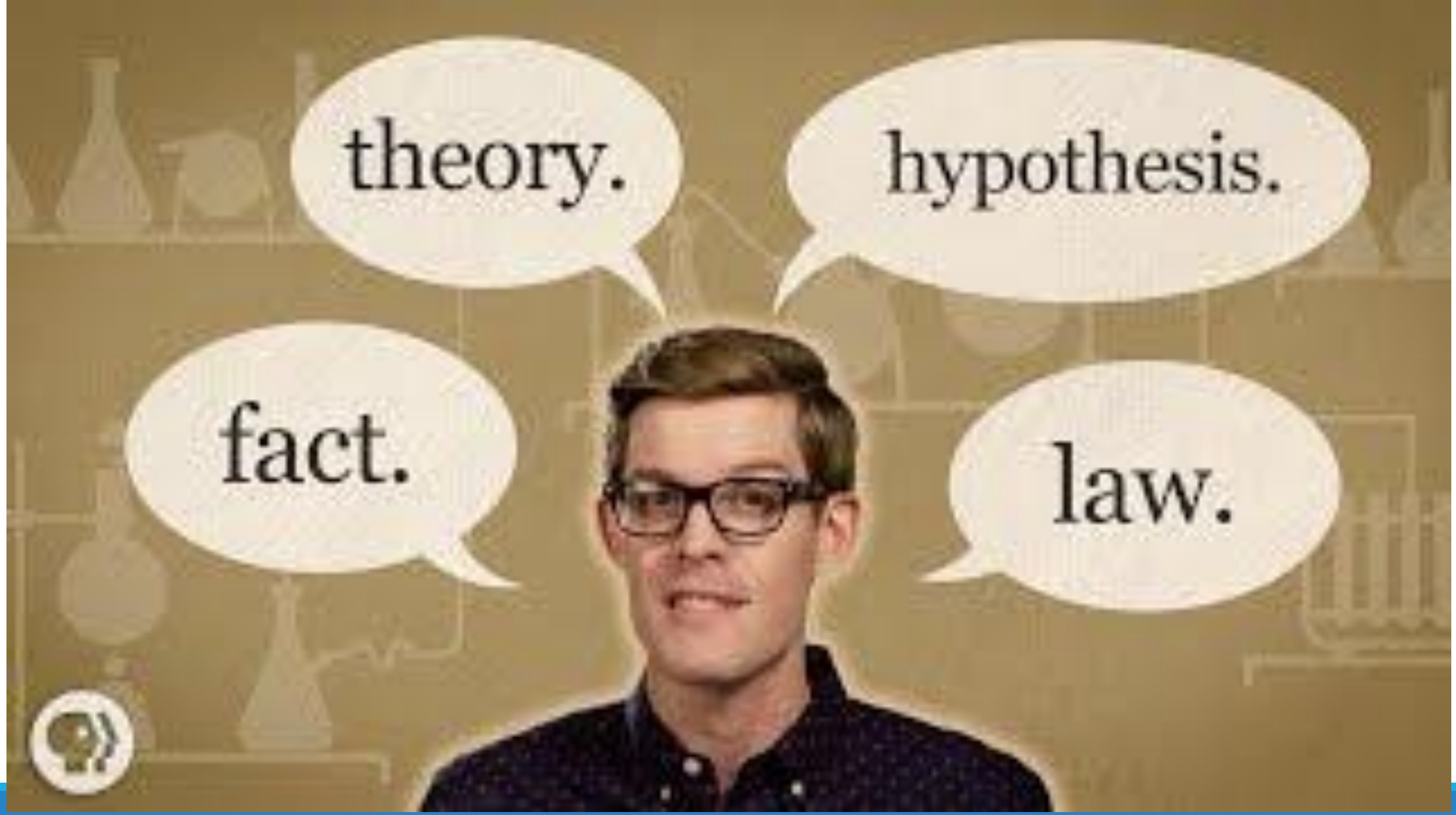
The Big Bang Theory & Beyond

What is cosmology?

- Cosmology is the scientific study of the **Universe**
- Cosmology involves the formation of **theories** or hypotheses that make specific predictions for phenomena that can be **tested** by observations
- Depending on the outcome of such observations, theories may be abandoned, **revised** or extended to accommodate new **data**


theory.
hypothesis.
fact.
law.

Science Talk - Fact vs Theory vs Hypothesis vs Law (video)

- Facts are just ___observations___ about the world around us.
- Hypothesis is a proposed ___explanation___ for a phenomenon made as a ___starting point___ for further investigation.
- When enough ___hypothesis___ have been tested ("given the old scientific checkmark") we can compile them and determine a ___theory___.

Science Talk - Fact vs Theory vs Hypothesis vs Law (video)

- A ___theory___ is a well-substantiated explanation acquired through the scientific method and repeatedly tested and confirmed through observation and experimentation.
- The best thing about a theory is that we can use it to make ___predictions___.

★ In science, being a theory isn't a bad thing!! It means that countless ★ experiments have been done and they all came to the same conclusion, which supports the theory.

Science Talk - Fact vs Theory vs Hypothesis vs Law (video)

- A ____law____ is a statement based on repeated experimental observations that describes some phenomenon of nature.
- A ____law____ does not describe why.

★ Science is never done, it's always changing! So how can we trust it? ★
The goal of science is to create frameworks to truly understand how things are the way they are now, so we can know how things will be in the future.

The Big Bang Theory

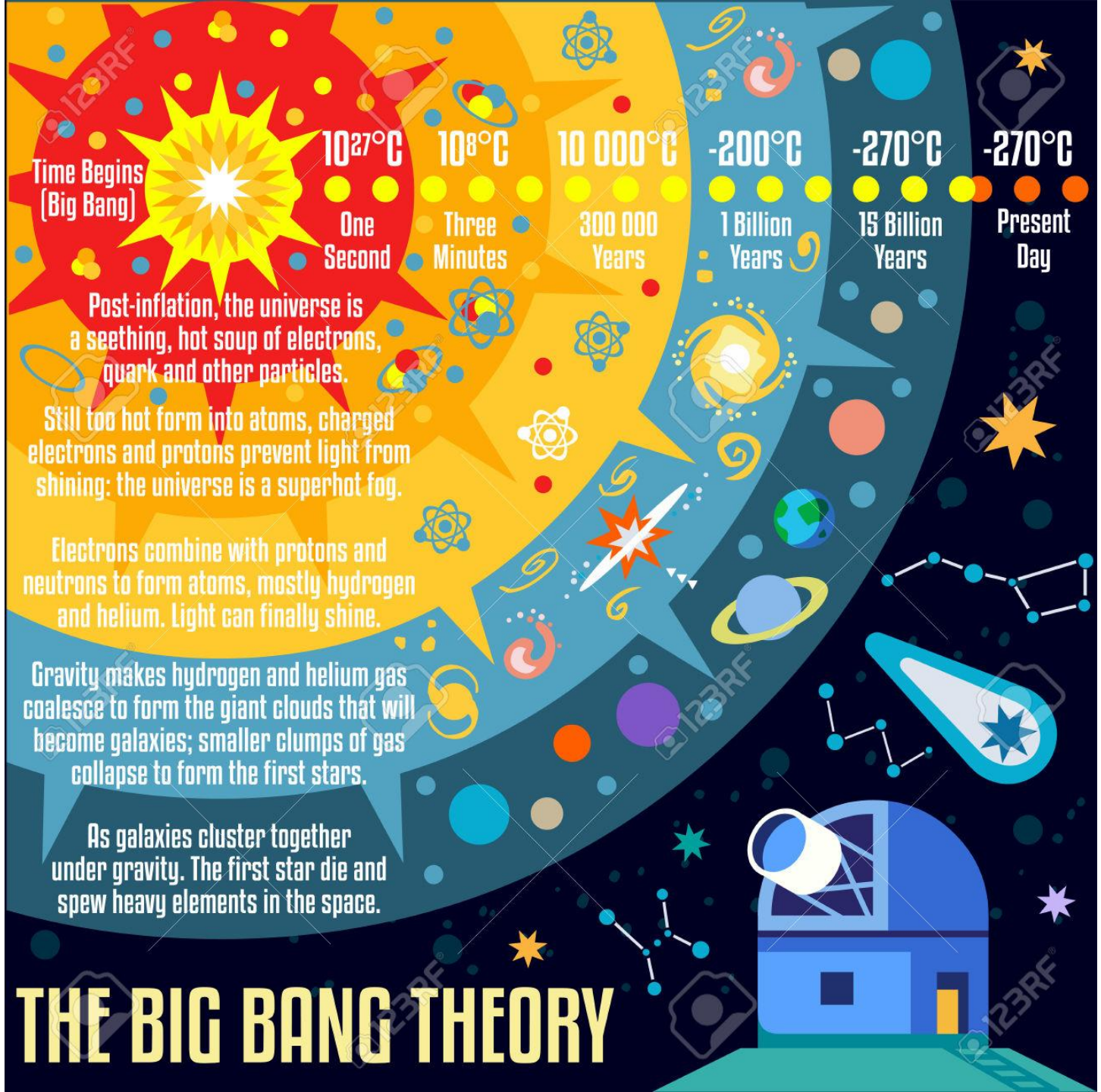
- **Model** for the origin and evolution of the Universe
- Around **14** billion years ago, everything in the Universe, including ***time*** and ***space***, existed in a single, fixed mass (singularity) that was hot and dense
- The mass then suddenly **expanded**

there was no down

The Big Bang Theory

- The Universe didn't expand into **anything** , because there wasn't **anything** for it to expand into
- The Universe expanded because the **space** between the galaxies expanded
- Everything, including **space** and **time** itself, was created in the Big Bang

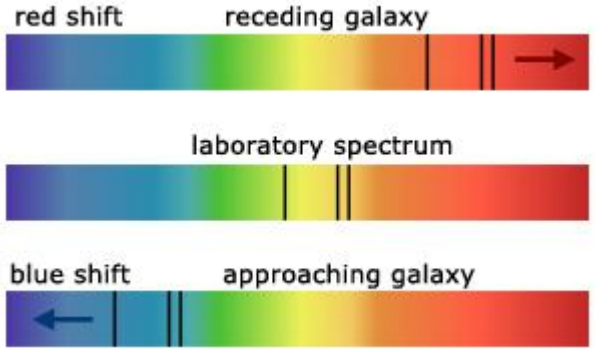
The Big Bang Theory

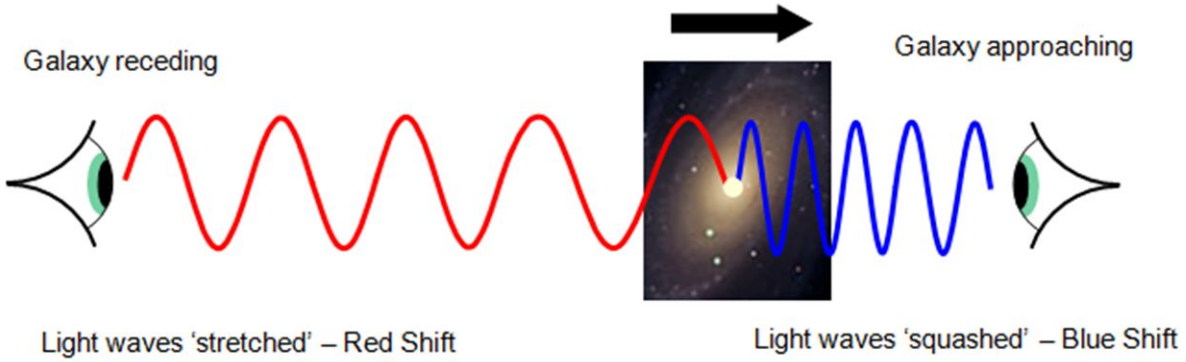


Three Pillars of Evidence

Red Shift

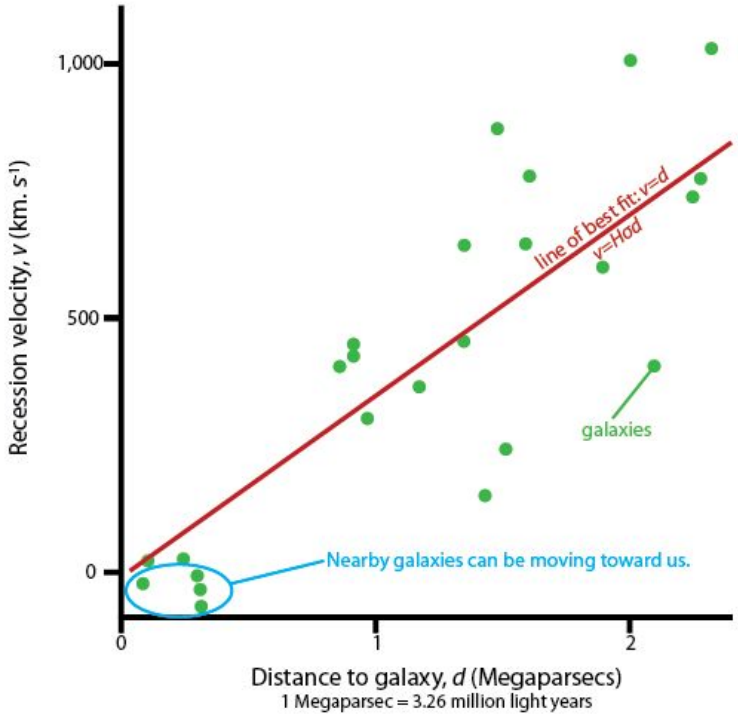
- Each galaxy emits its own distinctive **spectrum** of light
- Light spectra **shift**, depending on whether the light source is moving or stationary

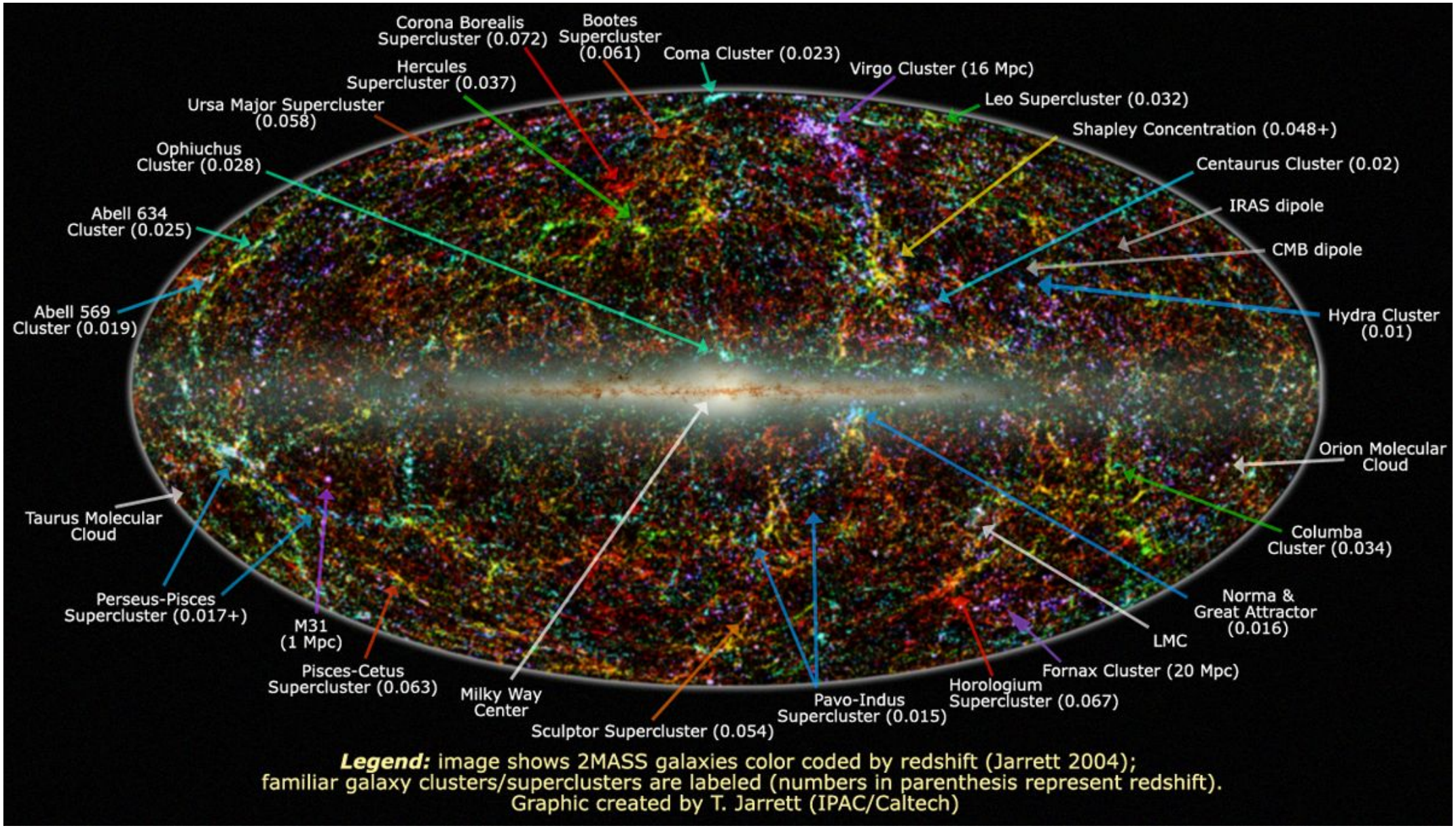




#1 The Universe is Expanding

- Spectroscopic red-shift data was used to measure the **speed** of distant objects then graphed their **distance** from Earth against their speed
- The farther away objects are from Earth, the **faster** they are moving away from us





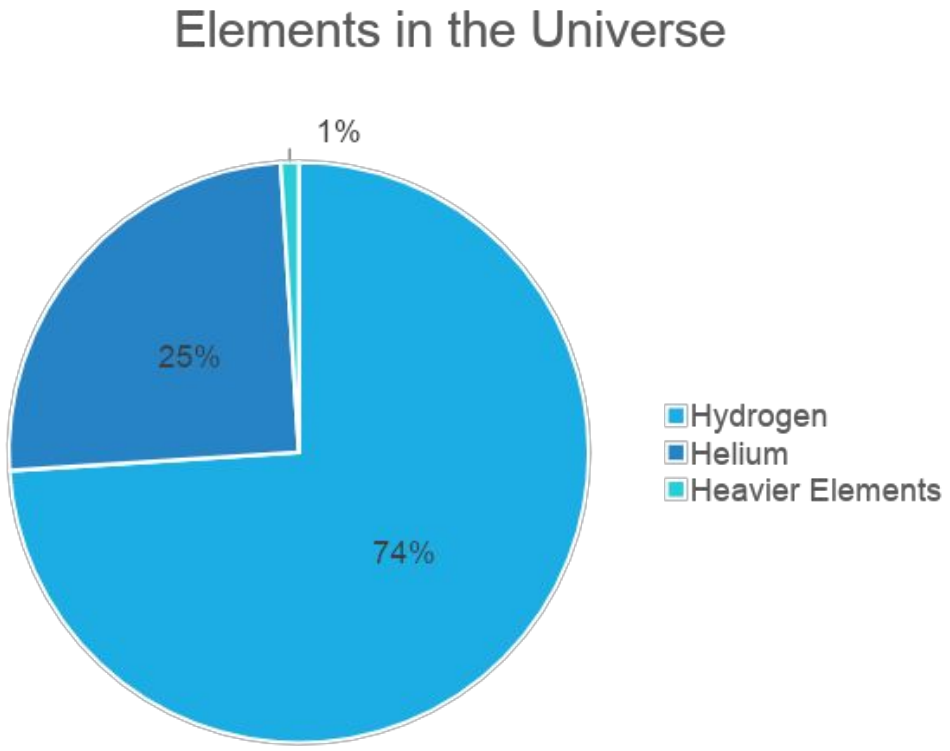
Legend: image shows 2MASS galaxies color coded by redshift (Jarrett 2004); familiar galaxy clusters/superclusters are labeled (numbers in parenthesis represent redshift). Graphic created by T. Jarrett (IPAC/Caltech)

Nucleosynthesis

- The Big Bang theory predicts that protons and neutrons fused together to form deuterium (an isotope of **hydrogen**) and **helium** in the first few minutes of the Universe
- Deuterium nuclei then combined to form **helium** and small amounts of **lithium**
- The predicted ratio would be about **3:1**; hydrogen to helium

#2 Abundance of H and He

- Using powerful telescopes, scientists have made extensive **spectroscopic** surveys of distant stars and galaxies
- What they have observed **matches** predictions made by the Big Bang Theory

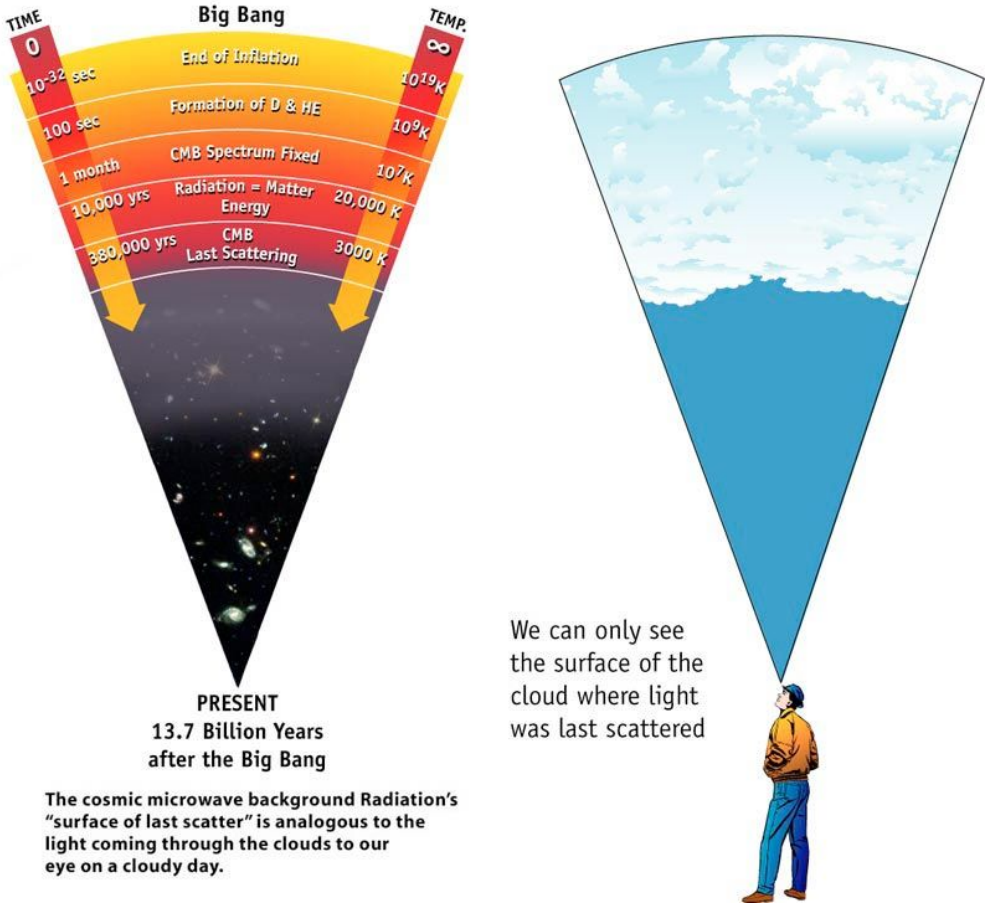


A Universe Full of Radiation!

- The Big Bang theory predicts that the early Universe was a very **hot** place and that as it expands, the gas within it **cools**
- Thus the universe should be filled with **radiation** that is literally the remnant **heat** left over from the Big Bang - cosmic microwave background radiation (CMBR)
- Using CMBR, cosmologists were able to calculate the theoretical temperature of today's Universe and began to search for **evidence** of it

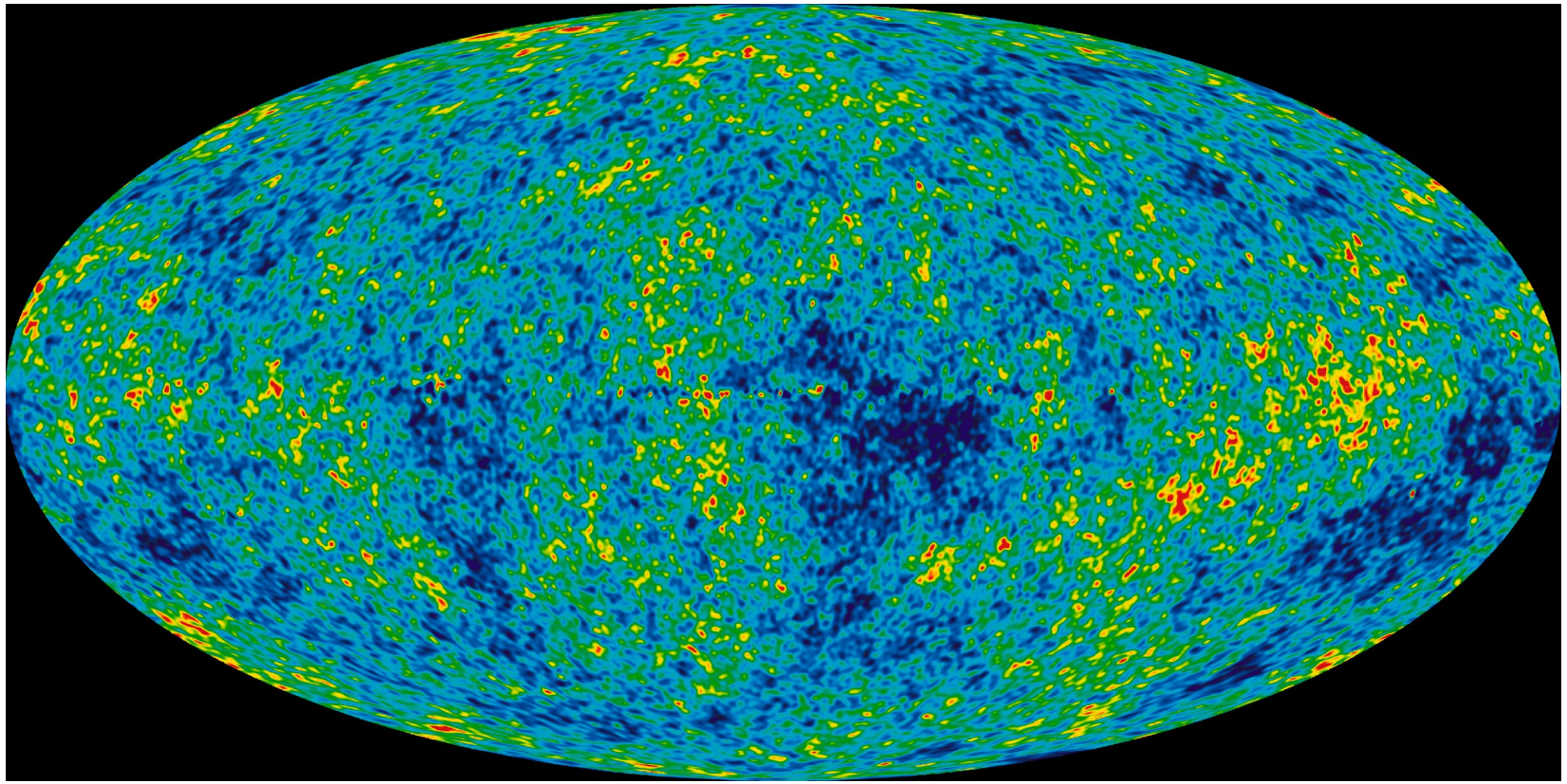
#3 CMBR

- It was eventually discovered by accident in **1964**
- This was the first **physical** evidence of the Big Bang



The cosmic microwave background Radiation's "surface of last scatter" is analogous to the light coming through the clouds to our eye on a cloudy day.

WMAP Survey



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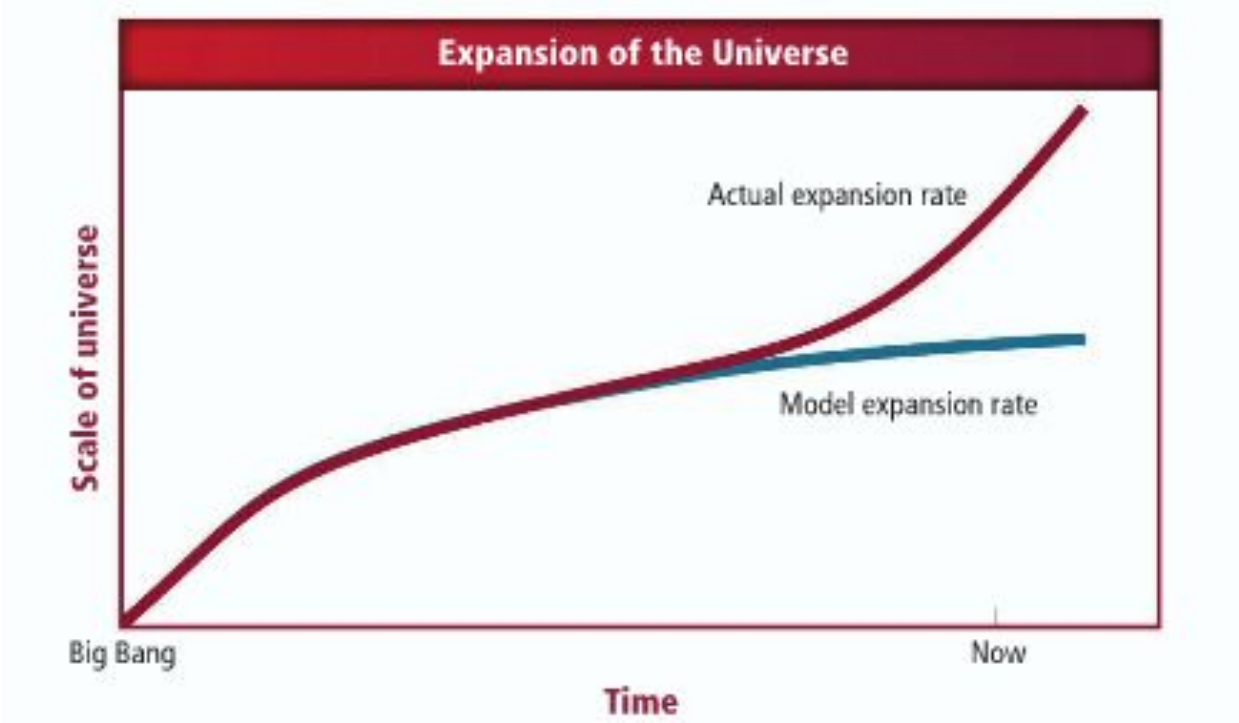
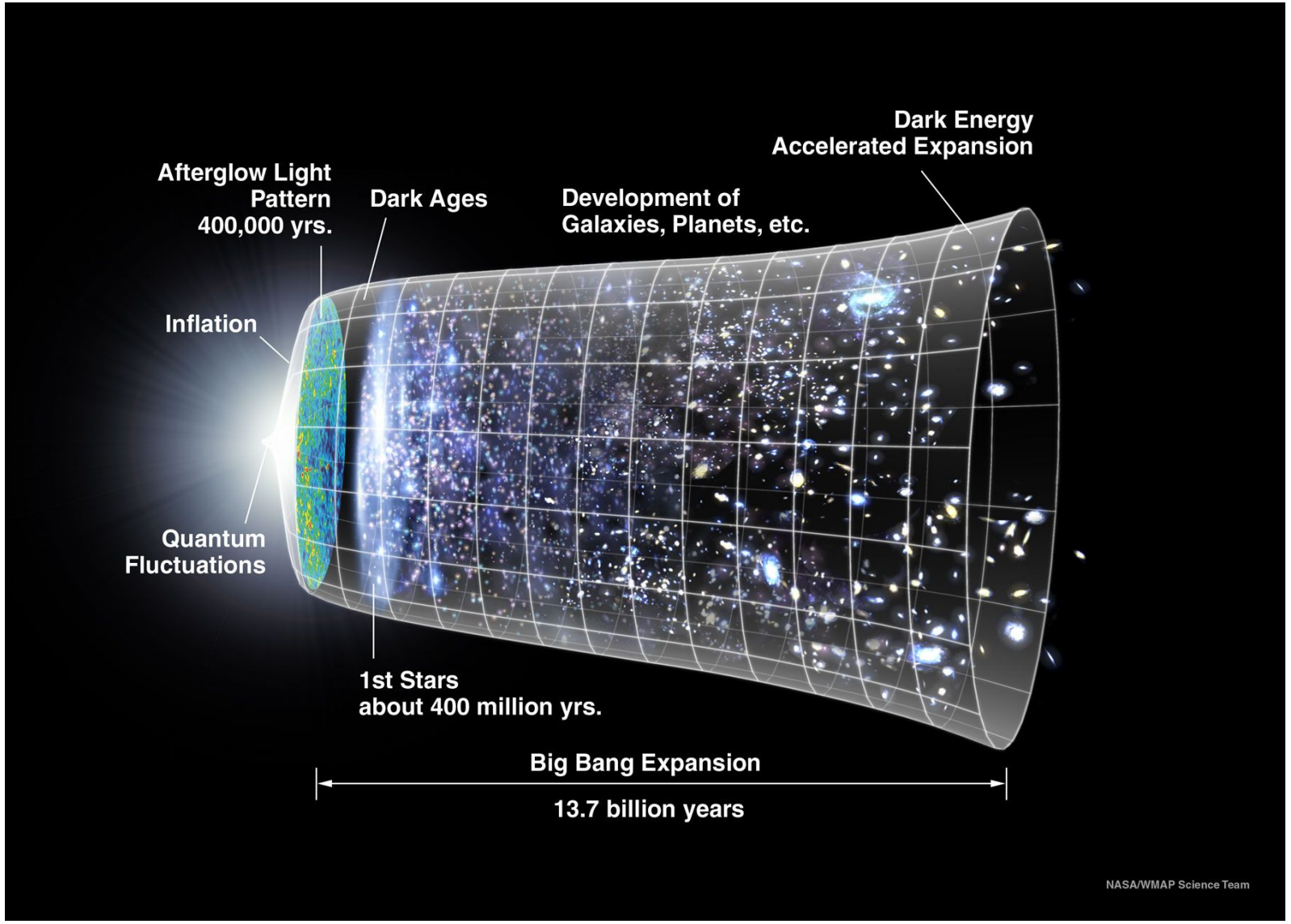
COSMOLOGY PART 1
THE BIG BANG

The Future

- According to every standard model, scientists thought that the expansion of the universe would **slow** down due to **gravity**
- In 2006, observations confirmed that the Universe was not only expanding – the expansion was **accelerating** !



Dark Energy


Dark Energy
Accelerated Expansion
Afterglow Light
Pattern
400,000 yrs.
Dark Ages
Development of
Galaxies, Planets, etc.
Inflation
Quantum
Fluctuations
1st Stars
about 400 million yrs.
Big Bang Expansion
13.7 billion years
NASA/WMAP Science Team

PBS
HOW THE
UNIVERSE
MIGHT END

How the Universe Might End (video)

1. What will the sun turn into at the end of its life? **A red giant**
2. How will this impact the Earth? **The oceans will boil away**
3. How will this impact the rest of the planets in our solar system?

   They will be scorched
4. Galaxies are expanding, but one is approaching us. What is the name of this galaxy? **Andromeda**
5. Why is this galaxy moving towards us? **Gravity, everything is always attracting everything else**

How the Universe Might End (video)

The Big Rip

a) Phantom dark energy would not only expand the space between things but expand things themselves. It wouldn't just push galaxies apart, it would push stars away from galaxies, and planets away from stars. Eventually even atoms and molecules and particles would rip apart. And in the final moments the very fabric of space would rip apart.

Vacuum Decay

b) The Higgs Field is an energy field that permeates all of space. All of space (atoms, molecules, planets, etc.) is held together by the Higgs field. However if the value of the Higgs Field were slightly different it could have a huge impact. In the early universe the Higgs field had a different value. There is a chance that a high energy event (high energy explosion or the evaporation of a black hole) could kick the Higgs Field to a lower value. This would create a new kind of space with different rules which would rip particles apart and swallow everything in the bubble and would ultimately collapse into a black hole.

How the Universe Might End (video)

Collapsing Universe

c) Space between galaxies is smaller and smaller, collisions would be more frequent, stars and planets would be flung out or orbit, giant black holes would merge, matter would whirl into the giant black holes and be superheated by friction so hot that it shoots out jets of radiation. Eventually all of the energy that's ever been emitted or exists in a black hole will be squeezed together into higher energy wavelengths and temperatures will rise until nuclear explosions rip stars and planets apart leaving space full of hot plasma. The universe will resemble the early moments after the big bang

The Heat Death

d) About 5 billion years ago the expansion of the universe started speeding up and it's still speeding up. The bigger our universe gets the faster it's expanding. Dark energy is pushing the expansion and the more space there is the more dark energy exists and the more it pushes. Gravity still exists holding solar systems and galaxies together, but the space between galaxies is dominated by dark energy. The universe is expanding faster than the speed of light. Eventually every galaxy will be alone in a dark universe. Stars will burn out and black holes will evaporate. There will be no fuel for new stars, just empty spaces filled with dark energy.